

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Theory Examination

December 2012

EE 101 Fundamentals of Electrical & Electronics Engineering (Old Course)

Date: 10.12.2012, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q – 1 (a) What do you understand by the term 'Resistance'? Also state and explain factors affecting resistance of a conductor. [04]
- (b) Explain effect of increase of temperature on resistance in case of (1) Metals, (2) Insulators and (3) Alloys. [03]
- Q – 2 (a) Explain the method of transforming a star connected network in a delta network and vice versa. [06]
- (b) Define following terms: [05]
- 1] Electric Field 2] Electric field strength 3] Relative permittivity
4] Electric field intensity 5] Dielectric strength
- (c) Calculate the force on a charge of $+10^{-2}$ C placed at a point (0, 0) due to point charges Q_1 and Q_2 . The charge $Q_1 = +10^{-6}$ C is at (-2, 2) meters and $Q_2 = -2 \times 10^{-6}$ C at (2, 2) meters. [03]

OR

- Q – 2 (a) Derive an expression for the energy stored in a capacitor of C farads when charged to a potential difference of V volts. [05]
- (b) A 20 μ F capacitor initially charged to a p.d. of 500 V is discharged through an unknown resistance. After one minute, the p.d. at the terminals of the capacitor is 200 V. what is the magnitude of the resistance? [05]
- (c) Explain Eddy current loss in detail. [04]
- Q – 3 (a) State and explain: (i) Fleming's Left hand (ii) Fleming's Right hand rule and (ii) Lenz's law. [06]
- (b) A coil is wound uniformly with 300 turns over a steel ring of relative permeability 900 having a mean diameter of 20 cm. The steel ring is made of a bar having cross-section of diameter 2 cm. If the coil has a resistance of 50 ohms and is connected to 250 V d.c. supply. Calculate (i) the m.m.f. (ii) the field intensity in the ring (iii) reluctance of the magnetic path (iv) total flux and (v) permeance of the ring. [05]
- (c) Define the followings: [03]
- [1] M.M.F. [2] Permeability [3] Magnetic Field Intensity

OR

- Q – 3 (a) Derive an expression for the instantaneous value of alternating sinusoidal emf in terms of its maximum value, angular frequency and time. [06]
- (b) Drive the relationship between line voltage and phase voltage, line current and phase current in case of 3 phase star connected system. [03]

- (c) A series RLC circuit with $R = 10 \Omega$, $L = 10 \text{ mH}$ and $C = 1 \mu\text{F}$ has an applied voltage of 200 V at resonant frequency. Calculate the resonant frequency, the current in the circuit and voltages across the elements at resonance. Find also the Q - factor and bandwidth. [05]

SECTION - II

- Q - 4 (a) (I) Explain any three terms: [06]
 (a) duty cycle, (b) fall time, (c) periodic signal, (4) positive logic system
 (II) Perform $(11110)_2 - (10010)_2$ using 2's complement.
 (b) Explain the operation of positive clamper. [03]
 (c) A crystal diode having internal resistance $r_f = 10 \Omega$ is used for half-wave rectification. If the applied voltage $v = 25 \sin \omega t$ and load resistance $R_L = 400 \Omega$ find: [03]
 (i) I_m, I_{dc}, I_{rms}
 (ii) DC output voltage
 (iii) Efficiency of rectification
- Q - 5 (a) Explain zener diode working as voltage stabilizer. [07]
 (b) Explain following Logic Gates in detail (i) OR gate, (ii) NAND gate. [02]
 (c) Give the advantages of digital over analog system. [02]

OR

- Q - 5 (a) Write in brief for following terms: [03]
 a. Knee voltage
 b. Forward current
 c. Peak Inverse Voltage
 (b) Convert $(1427)_8$ into hexa-decimal number. [02]
 (c) (i) Convert $(1110010111011)_2$ number into Hexa decimal number. [06]
 (ii) Convert $(A2E7)_{16}$ number into Octal number.
 Q - 6 (a) Derive the relationship between α and β . [04]
 (b) Name the three possible transistor configurations. Also show that α is always less than unity. [04]
 (c) Describe the transistor action in detail. [04]

OR

- Q - 6 (a) (i) Give in brief about operating point. [06]
 (ii) Calculate emitter current in a transistor for which $\beta = 40$ & base current = $15 \mu\text{A}$
 (iii) find the α if (a) $\beta = 90$, (b) $\beta = 40$
 (b) Explain common emitter configuration in detail and also derive the useful equations. [06]
